

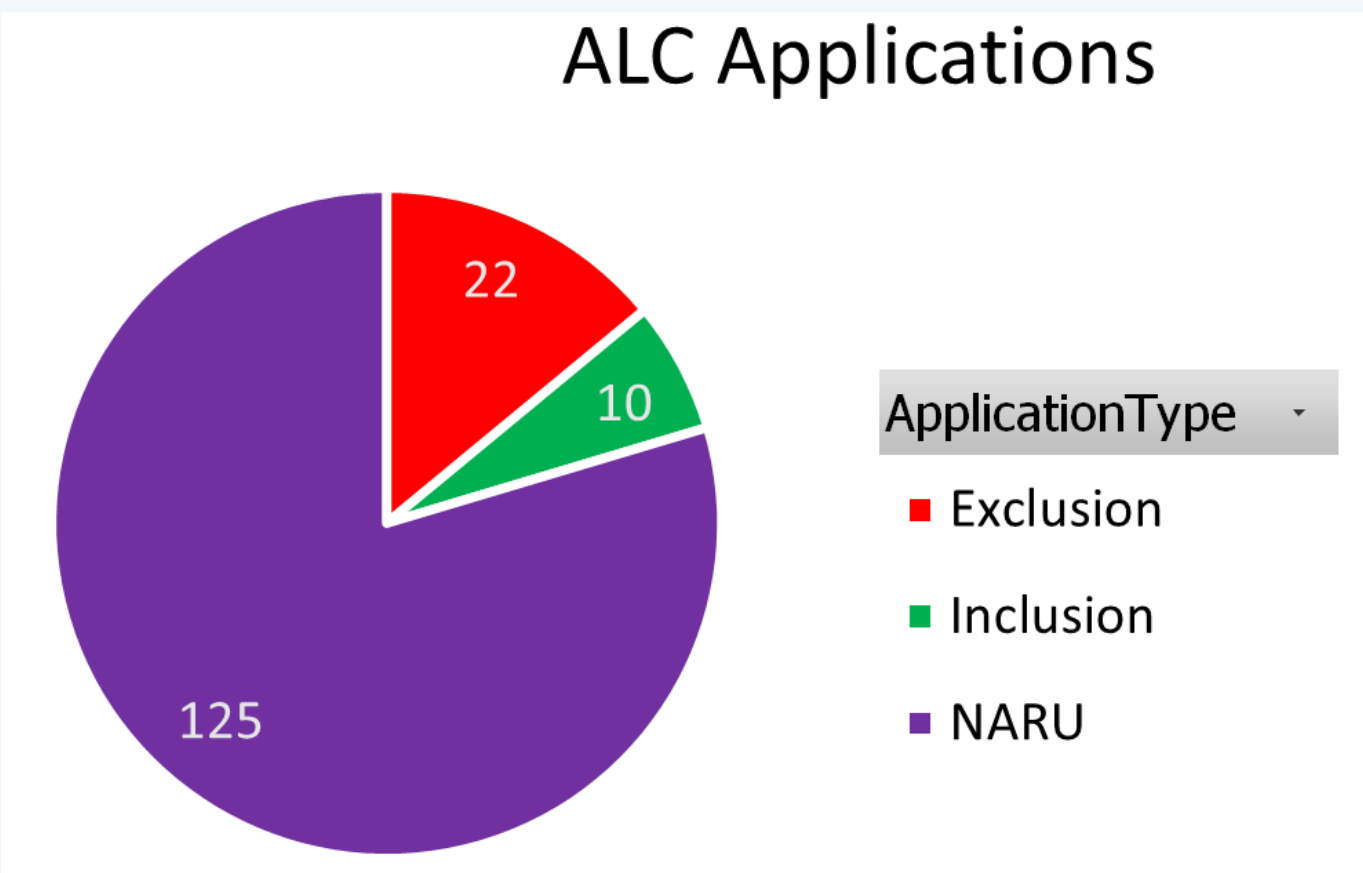
Land Lost to ALC Applications in Greater Vancouver

Brandon Fleming

Kwantlen Polytechnic University

Introduction

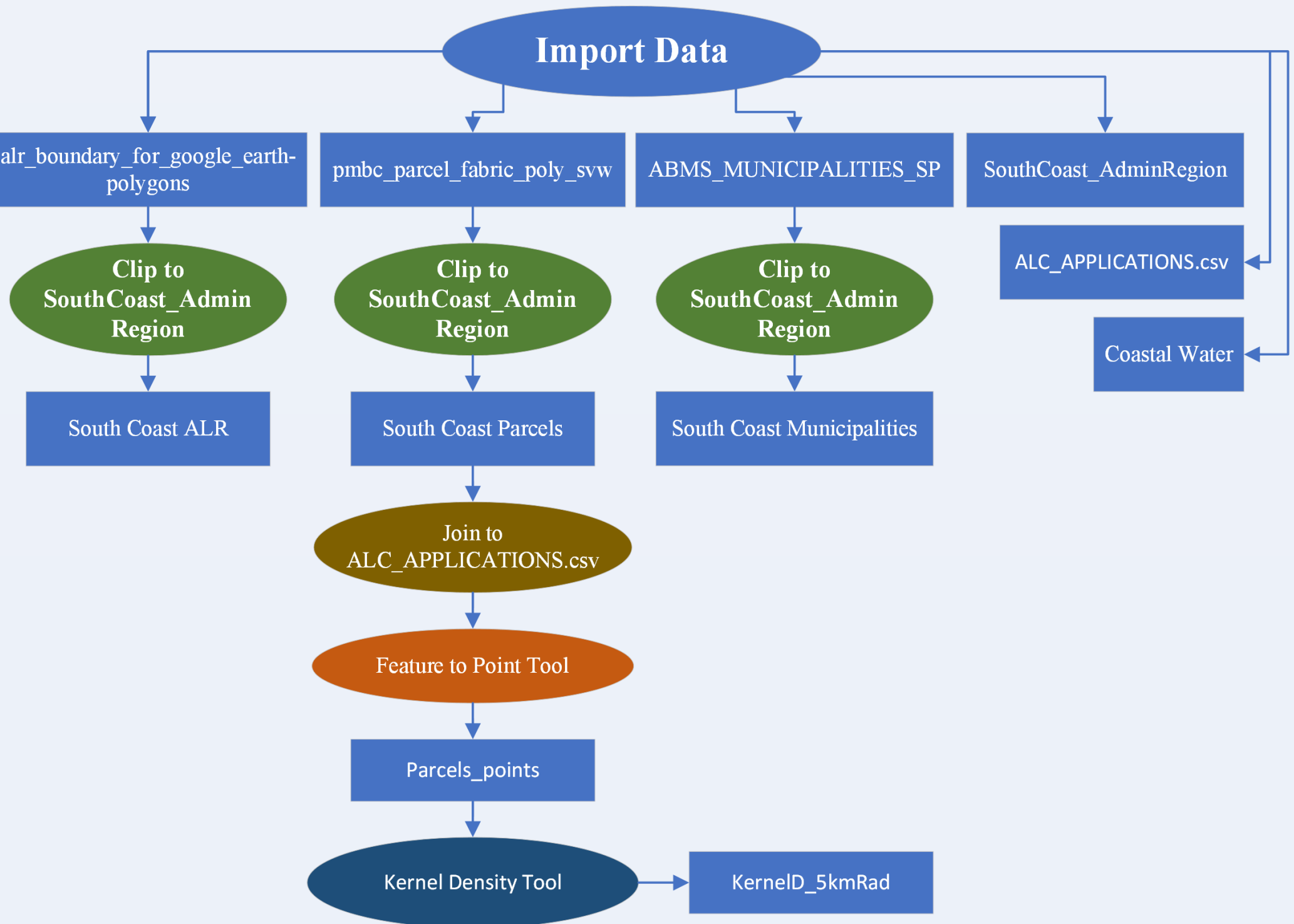
The preservation of agricultural land is integral to ensuring the long-term vitality of our communities and the environment. With the recent push to build more and more homes, the agricultural land is feeling the squeeze. In this poster I delve into the Agricultural Land Reserve (ALR) as well as the organization which protects it, the Agricultural Land Commission (ALC). I investigate the loss of agricultural land through different types of approved applications: inclusion, exclusion, and non-adhering residential use (NARU). I chose this topic because I just finished an internship at the ALC and feel very passionate about protecting farmland.



Methodology

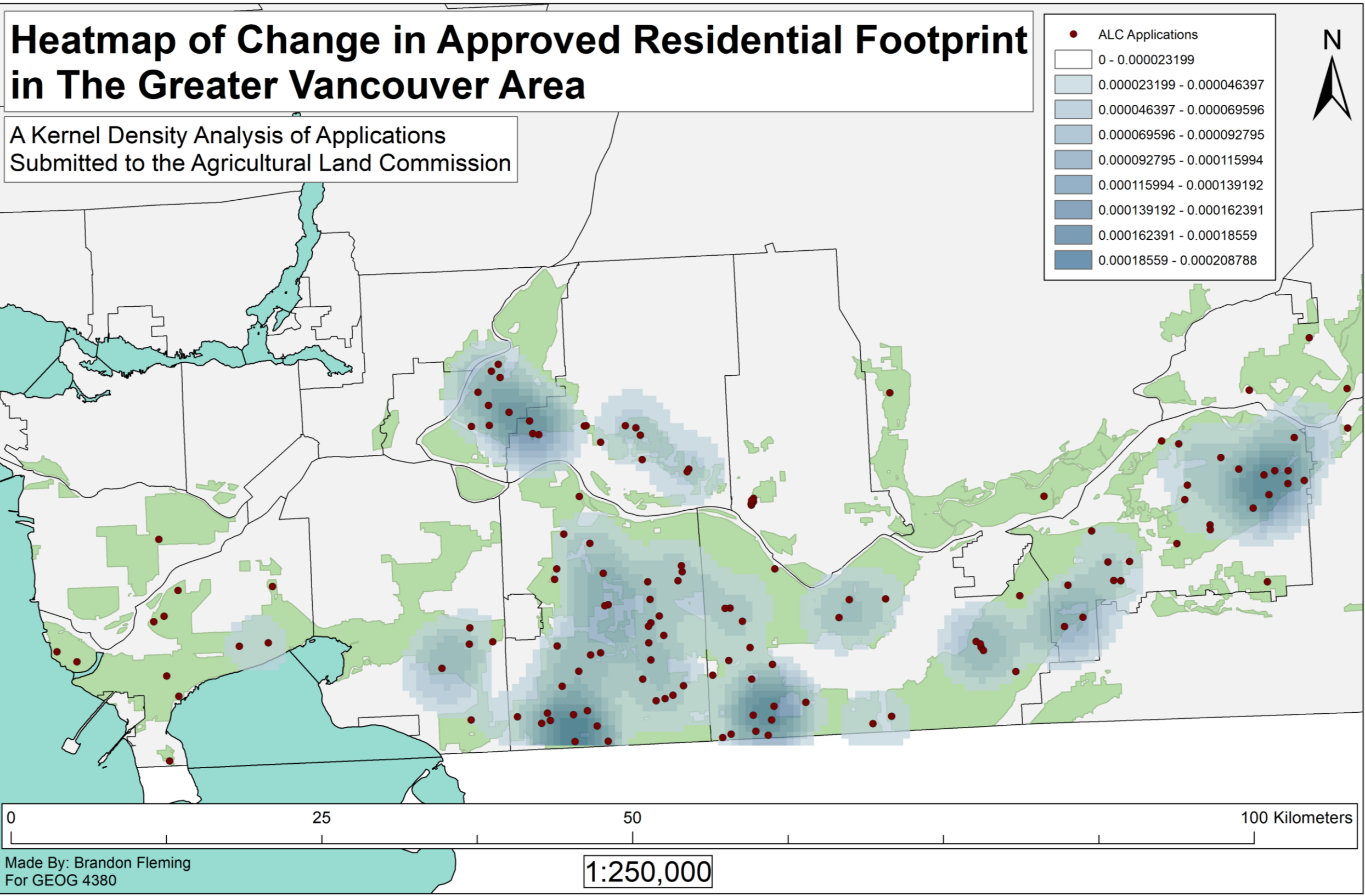
Data Collection

I used publicly available application information from the ALC to figure out how much land has been lost to the approved applications. Using the ALC’s application portal, I was able to build my dataset. I gathered information on date, parcel ID, application type, land lost, land gained, and change in total residential footprint. For some of the NARU applications I was also able to find the existing house size. After importing the .csv file with my data on it I had to export the data before I was able to create a new field, make it long integer, and select “PID” as the input. This was the only way to transform the data from string to numerical.



Scope of Data

I chose to limit the geographical scope of the data to encompass the Greater Vancouver Area. I chose this region because it faces the most pressure to remove agricultural land from the reserve. The temporal scope was decided to be different based on the application type. The inclusion and exclusion applications I chose to go as far back as 2017 because that is as far back as the ALC portal goes. With the Non-adhering residential use applications (NARU) I was only able to go as far back as 2019 because that is when the ALC first started receiving that type of application. I also narrowed the scope for application type. I chose exclusion, inclusion, and NARU because they are the most straightforward and deal with and all dealt with a change in ALR land. Other types such as “transportation/utility/trail” might be interesting to look at in a future project but do not always deal with a change in ALR.



Process

I first imported each layer into ArcMap. Then I clipped the ALR land layer, the BC parcel map layer, and the municipalities layer to the South Coast layer. Next, I imported my spreadsheet .csv file and joined it to the BC parcel map layer. The common feature I used to join the data was parcel identifier (PID). For the density analysis map (above), I used the “Feature to Point” tool to turn the polygons into points so that I could use the “Kernel Density” tool in ArcMap and set the radius to 5 square kilometers. The kernel density was tied to the “change in residential footprint” feature. In the application type map (to the right), I symbolized the data by application type (exclusion, inclusion, or NARU). For the application type I chose to make the inclusion green, the exclusion red, and the NARU purple.

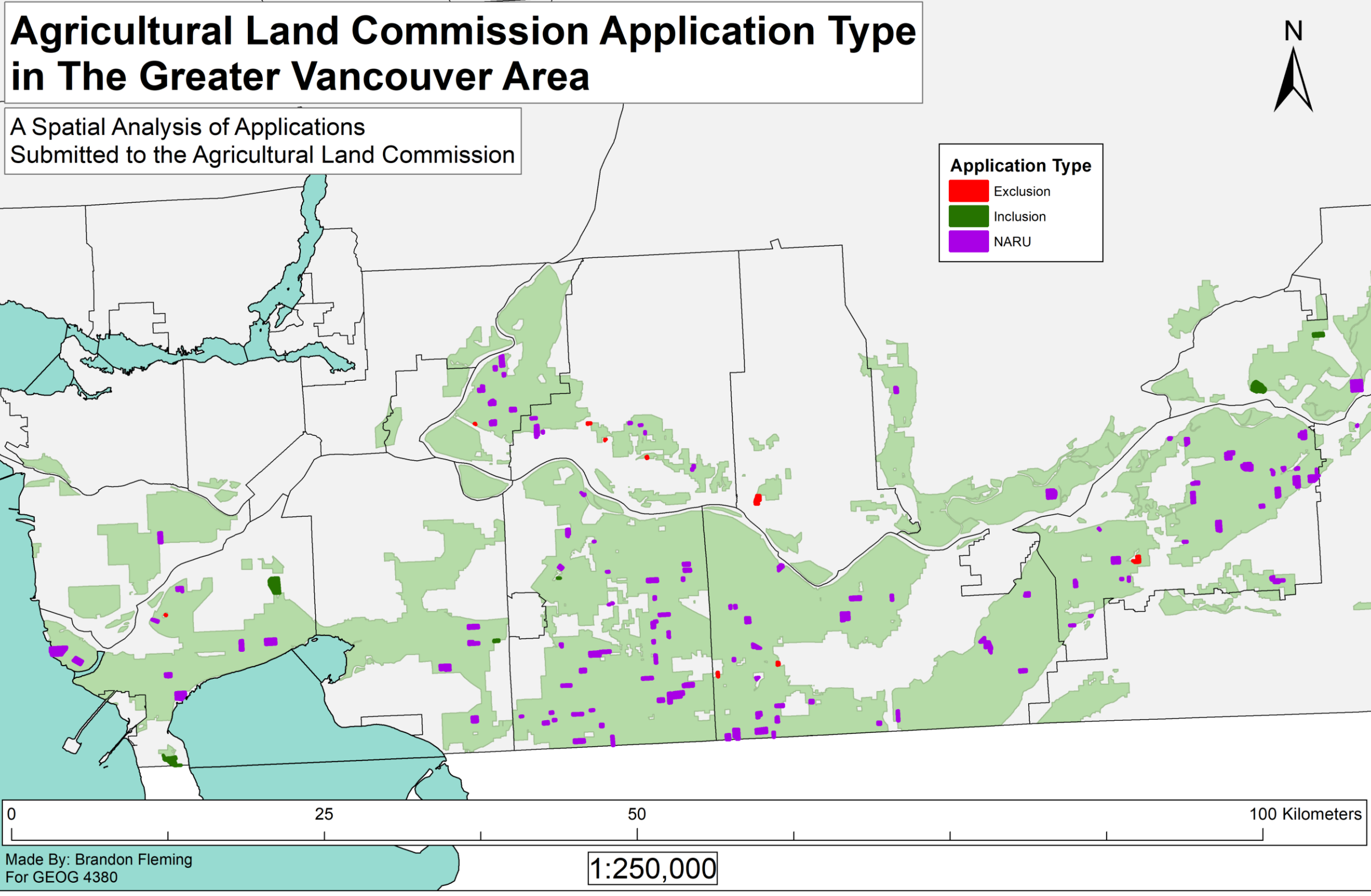
Findings

Overall Patterns and Trends

The total amount of land lost to residential buildings on ALR is about 67,000m². The total amount of land gained from inclusion and exclusion applications is 1,890,000m². Looking first at the application type map we can see that most of the exclusion applications are on the edge of communities. When I was sifting through the application data, I noticed a lot of these are lots that were quickly subdivided into more dense residential land once they had been excluded from the ALR. This is what I expected to see. The inclusion applications are the same in this regard. The NARU applications hotspots are closer to more populous areas like Chilliwack or Aldergrove but are also almost nonexistent in Richmond and Surrey. What is the most interesting about this map to me is the lack of applications within Surrey, which has nearly half of its land in the ALR and has a quickly growing population. At the same time the Township of Langley has by far the most applications and more than one hotspot. I believe this is due to the restrictions each municipality has in place. Township of Langley is more relaxed while Richmond is stricter.

Using another database, one with existing home size data collected, I ran the average size of existing and approved homes within NARU applications. I found that the existing homes were on average about 170 square meters while the newer applied for and approved homes were an average of 400 square meters (right top). A time-based analysis done with the NARU applications shows an average increase of about 200 plus square meters from 2017 to 2023 (right bottom).

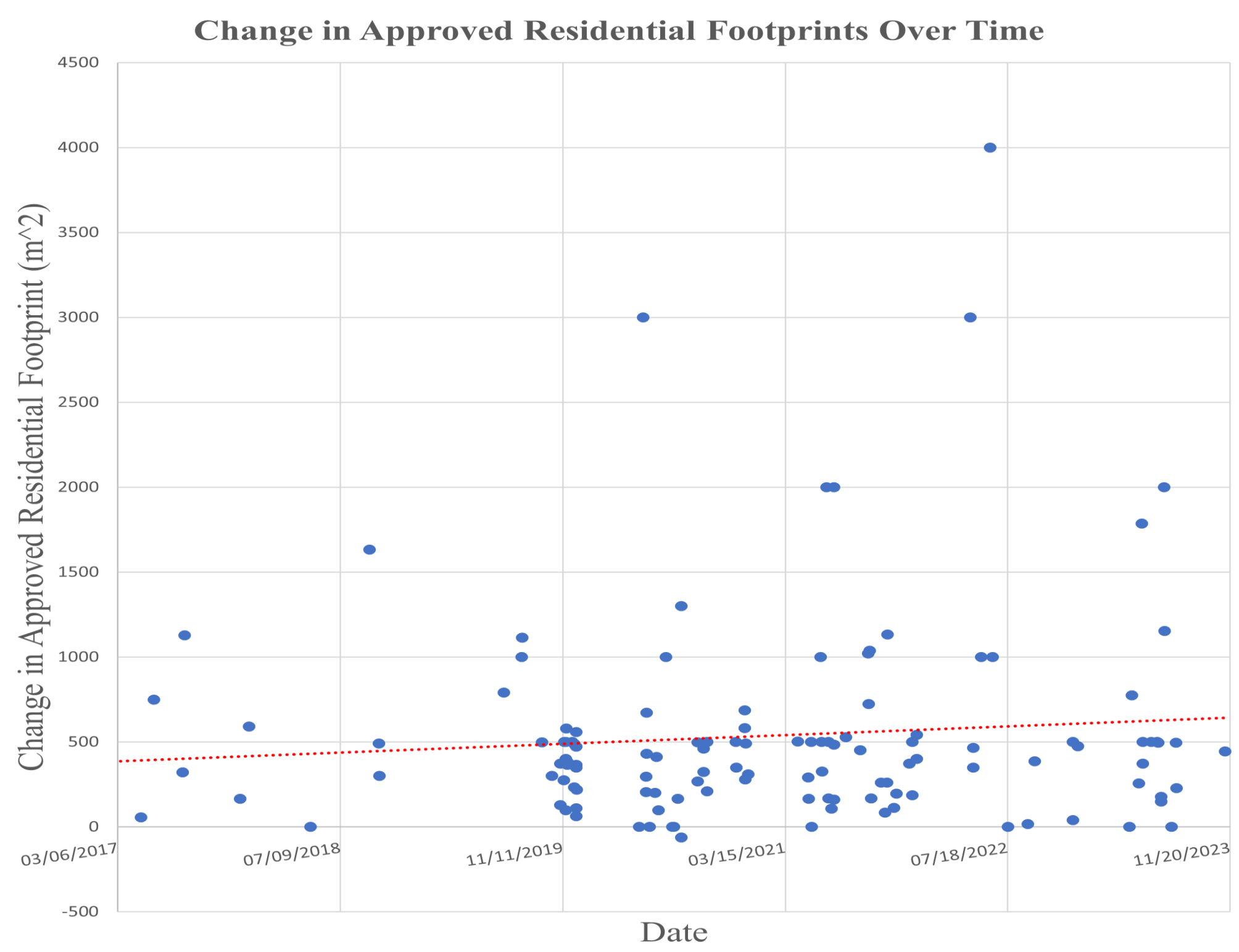
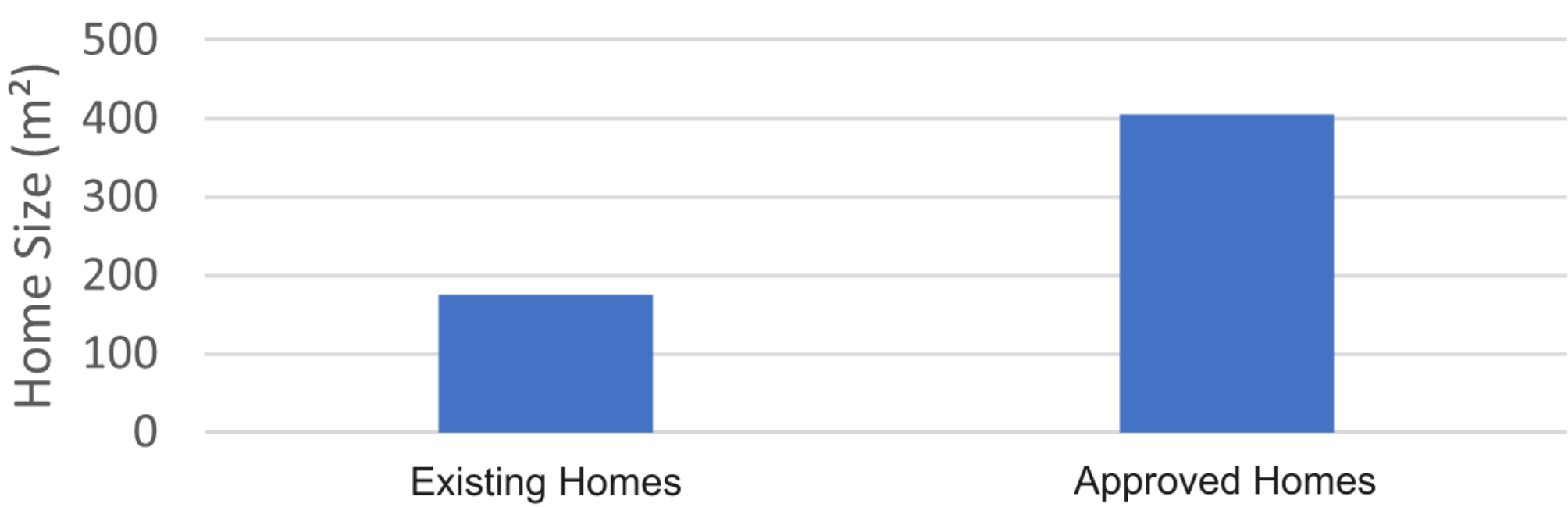
In future research I might open the geographical scope to all of BC or look at applications older than 2017 if I can get access to them.



Conclusion

Clearly, the loss of land is not due to exclusion applications (which are more than balanced out with inclusion) but rather the NARU applications. Despite taking small bites out of the ALR, there is a loss of land. The average approved home size is increasing. The proposed home is on average 230% larger than the already existing home. Not all these properties with home increases will have a proportional increase in agricultural output. Stricter regulations on building new homes on ALR should be implemented. As we walk the fine line between urban development and agricultural preservation, we have to understand the impact of the ALC’s NARU applications. This poster is a good start.

Average Home Size Change After NARU Application



References

- Agricultural Land Reserve Maps—Provincial Agricultural Land Commission. (2022, February 8). <https://www.alc.gov.bc.ca/alr-maps/>
- Agricultural Land Commission. (n.d.). *ALR Portal—Welcome*. Retrieved October 13, 2023, from <https://a100.gov.bc.ca/pub/oatsp/list?execution=e2s1>
- Government of British Columbia. (2023, November 20). *ParcelMap BC Parcel Fabric—Datasets—Data Catalogue*. <https://catalogue.data.gov.bc.ca/dataset/parcelmap-bc-parcel-fabric>

Acknowledgements

I would like to acknowledge Katie Lambert of the ALC GIS team.

December 3rd, 2023